Import Data

In [2]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

In [3]:
```
data = pd.read_csv(r"D:\Downloads\Udemy-PY\11. airlines_flights_data.csv")
```

In [6]:
```
data
```

Out[6]:

	index	airline	flight	source_city	departure_time	stops	arrival_time	destina
0	0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	
1	1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	
2	2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	
3	3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	
4	4	Vistara	UK-963	Delhi	Morning	zero	Morning	
...	...	...	...	...	...	...	...	
300148	300148	Vistara	UK-822	Chennai	Morning	one	Evening	Hy
300149	300149	Vistara	UK-826	Chennai	Afternoon	one	Night	Hy
300150	300150	Vistara	UK-832	Chennai	Early_Morning	one	Night	Hy
300151	300151	Vistara	UK-828	Chennai	Early_Morning	one	Evening	Hy
300152	300152	Vistara	UK-822	Chennai	Morning	one	Evening	Hy

300153 rows × 12 columns

Data Cleaning

In [9]:
```
data.drop( columns = 'index', inplace = True)
```

In [10]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300153 entries, 0 to 300152
Data columns (total 11 columns):
 #   Column            Non-Null Count   Dtype  
---  ------            --------------   -----  
 0   airline           300153 non-null  object 
 1   flight            300153 non-null  object 
 2   source_city       300153 non-null  object 
 3   departure_time    300153 non-null  object 
 4   stops             300153 non-null  object 
 5   arrival_time      300153 non-null  object 
 6   destination_city  300153 non-null  object 
 7   class             300153 non-null  object 
 8   duration          300153 non-null  float64
 9   days_left         300153 non-null  int64  
 10  price             300153 non-null  int64  
dtypes: float64(1), int64(2), object(8)
memory usage: 25.2+ MB
```

In [11]:
```
data.describe()
```

Out[11]:

	duration	days_left	price
count	300153.000000	300153.000000	300153.000000
mean	12.221021	26.004751	20889.660523
std	7.191997	13.561004	22697.767366
min	0.830000	1.000000	1105.000000
25%	6.830000	15.000000	4783.000000
50%	11.250000	26.000000	7425.000000
75%	16.170000	38.000000	42521.000000
max	49.830000	49.000000	123071.000000

In [13]:
```
data[data['duration']== 49.830000 ]                    # Flight with the max Duratio
```

Out[13]:

	airline	flight	source_city	departure_time	stops	arrival_time	destinatior
193889	Air_India	AI-672	Chennai	Evening	two_or_more	Evening	Banç
194359	Air_India	AI-672	Chennai	Evening	one	Evening	Banç

1. What are the Flights in the dataset, accompanied by their Frequencies?

In [17]:
```
data.head(2)
```

Out[17]:

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	cla
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Econor
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Econor

In [18]:
```
data['airline'].unique()
```

Out[18]:
```
array(['SpiceJet', 'AirAsia', 'Vistara', 'GO_FIRST', 'Indigo',
       'Air_India'], dtype=object)
```

In [20]:
```
data['airline'].value_counts()                    # All flights with their frequencies
```

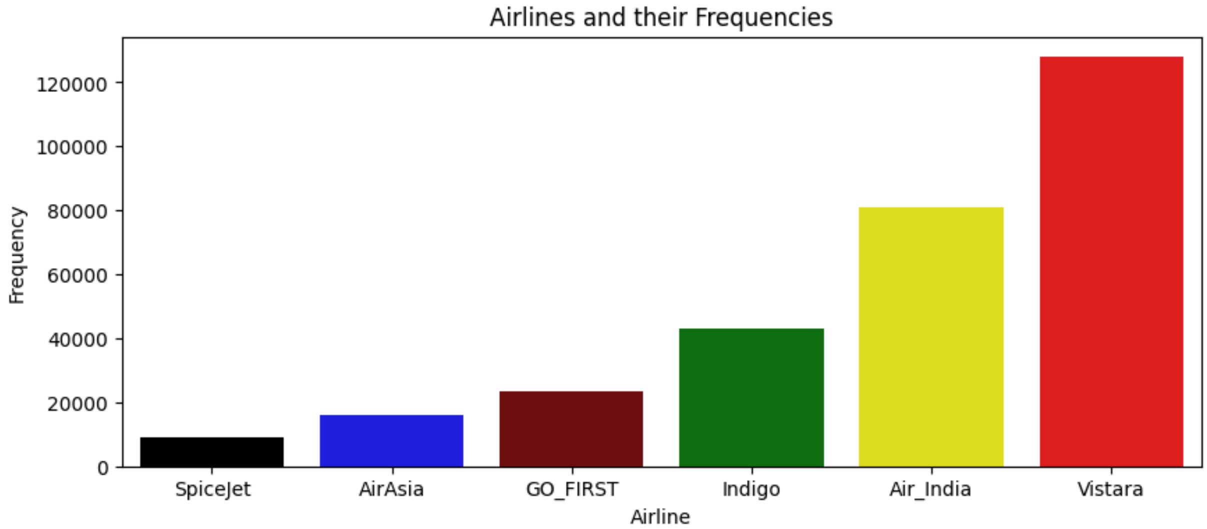
Out[20]:
```
airline
Vistara      127859
Air_India     80892
Indigo        43120
GO_FIRST      23173
AirAsia       16098
SpiceJet       9011
Name: count, dtype: int64
```

In [34]:
```
plt.figure(figsize = (10,4))
sns.barplot(data['airline'].value_counts(ascending = True), palette = ['black', 'bl
plt.title('Airlines and their Frequencies')
plt.xlabel('Airline')
plt.ylabel('Frequency')
plt.show();
```

```
C:\Users\hmmuz\AppData\Local\Temp\ipykernel_22872\2482389260.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data['airline'].value_counts(ascending = True), palette = ['black', 'b
lue', 'maroon', 'green', 'yellow', 'red'])
```



2. Draw a bar graph to show Depature time and Arrival time

In [40]:
```
data['departure_time'].value_counts()
```

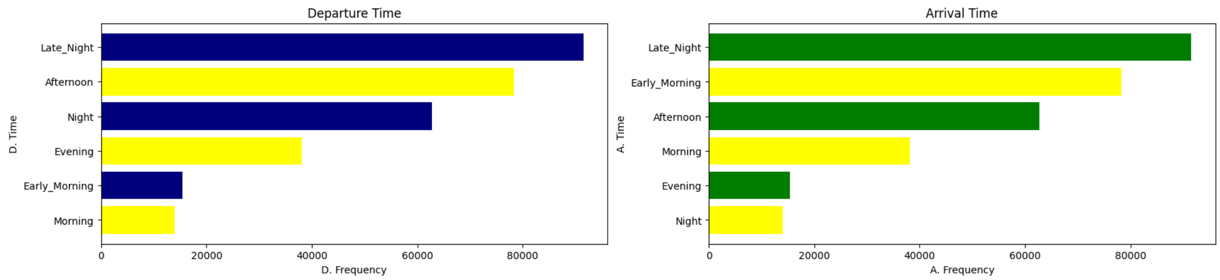
Out[40]:
```
departure_time
Morning          71146
Early_Morning    66790
Evening          65102
Night            48015
Afternoon        47794
Late_Night        1306
Name: count, dtype: int64
```

In [42]:
```
data['arrival_time'].value_counts()
```

Out[42]:
```
arrival_time
Night            91538
Evening          78323
Morning          62735
Afternoon        38139
Early_Morning    15417
Late_Night       14001
Name: count, dtype: int64
```

In [69]:
```
plt.figure(figsize = (20,4))
plt.subplot (1,2,1)
plt.barh(data['departure_time'].value_counts().index , data['arrival_time'].value_c
plt.title('Departure Time')
plt.xlabel('D. Frequency' )
plt.ylabel('D. Time')

plt.subplot(1,2,2)
plt.barh(data['arrival_time'].value_counts().index, data['arrival_time'].value_coun
plt.title('Arrival Time')
plt.xlabel('A. Frequency')
plt.ylabel('A. Time')
plt.show();
```



3. Draw a bar gragh showing the Source City and Destination City

In [70]:
```
data.head(2)
```

Out[70]:

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	cla
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Econor
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Econor

In [76]:
```
data['source_city'].value_counts()
```

Out[76]:
```
source_city
Delhi        61343
Mumbai       60896
Bangalore    52061
Kolkata      46347
Hyderabad    40806
Chennai      38700
Name: count, dtype: int64
```

In [78]:
```
data['destination_city'].value_counts()
```

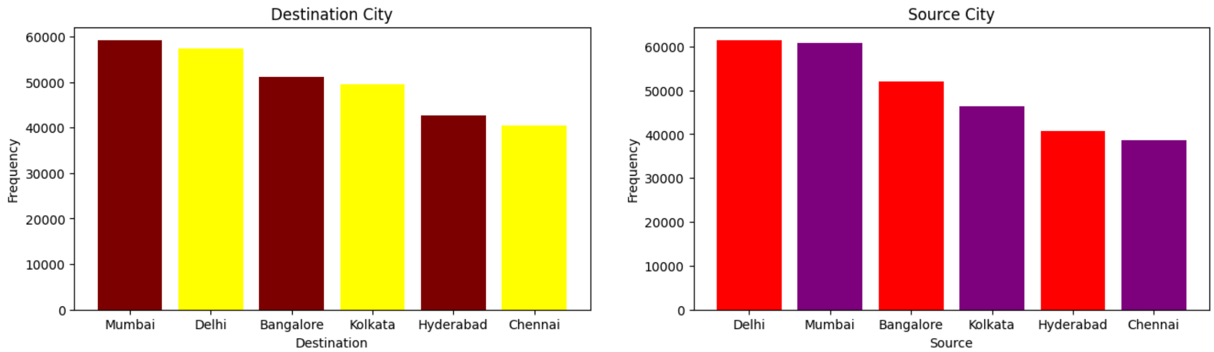
Out[78]:
```
destination_city
Mumbai       59097
Delhi        57360
Bangalore    51068
Kolkata      49534
Hyderabad    42726
Chennai      40368
Name: count, dtype: int64
```

In [83]:
```
plt.figure(figsize = (16,4))
plt.subplot(1,2,2)
plt.bar(data['source_city'].value_counts().index, data['source_city'].value_counts(
plt.title('Source City')
plt.xlabel('Source')
plt.ylabel('Frequency')

plt.subplot(1,2,1)
```

```
plt.bar(data['destination_city'].value_counts().index, data['destination_city'].val
plt.title('Destination City')
plt.xlabel('Destination')
plt.ylabel('Frequency')

plt.show();
```



4. Does Prices varry with the Airlines?

In [85]:
```
data.head(2)
```

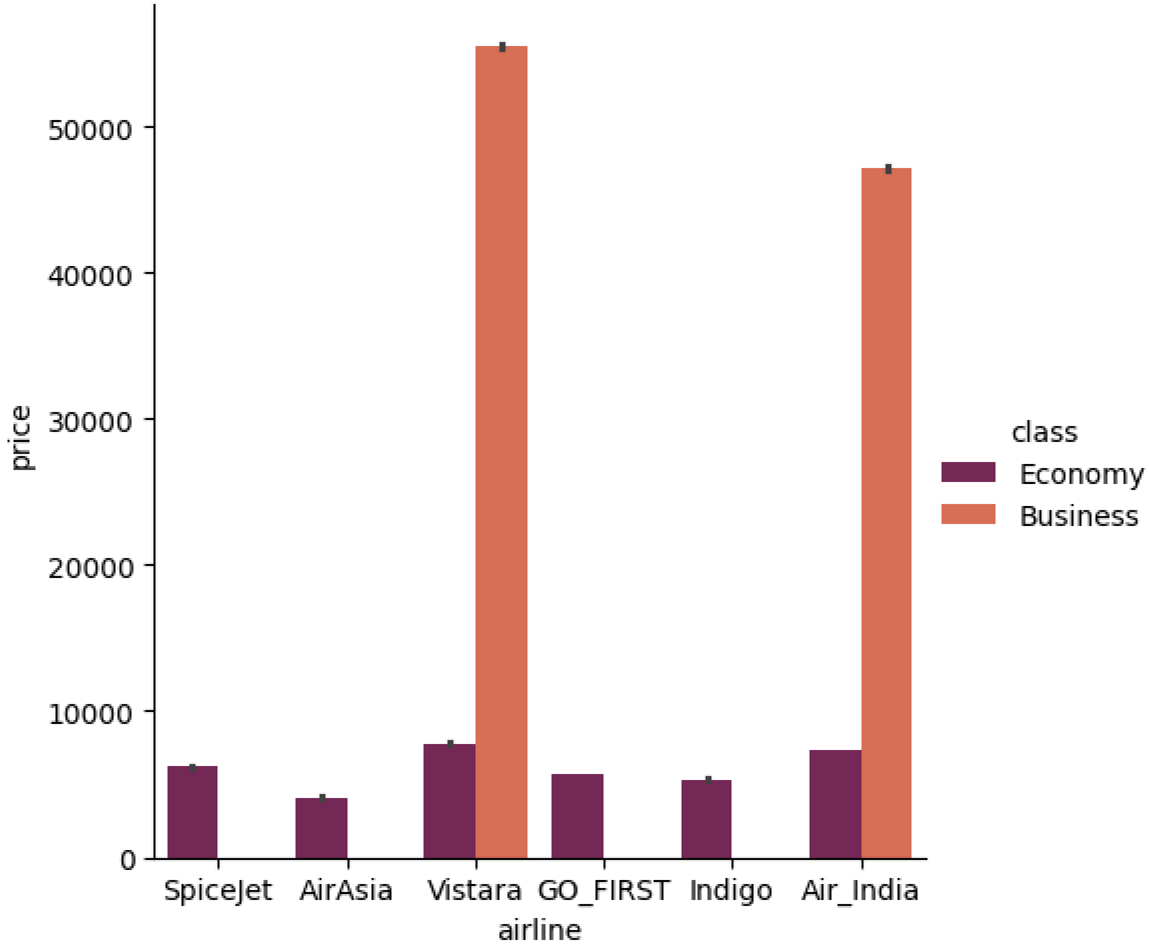
Out[85]:

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	cla
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Econor
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Econor

In [86]:
```
data.groupby('airline')['price'].mean()
```

Out[86]:
```
airline
AirAsia       4091.072742
Air_India    23507.019112
GO_FIRST      5652.007595
Indigo        5324.216303
SpiceJet      6179.278881
Vistara      30396.536302
Name: price, dtype: float64
```

In [88]:
```
sns.catplot(x= 'airline', y = 'price',kind = 'bar', data = data, hue = 'class', pal
plt.show()
```



5. Does Ticket price changes with Departure time and Arrival time?

In [89]:
```
data.head(2)
```

Out[89]:

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	cla
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Econor
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Econor

In [93]:
```
data.groupby('departure_time')['price'].mean()
```

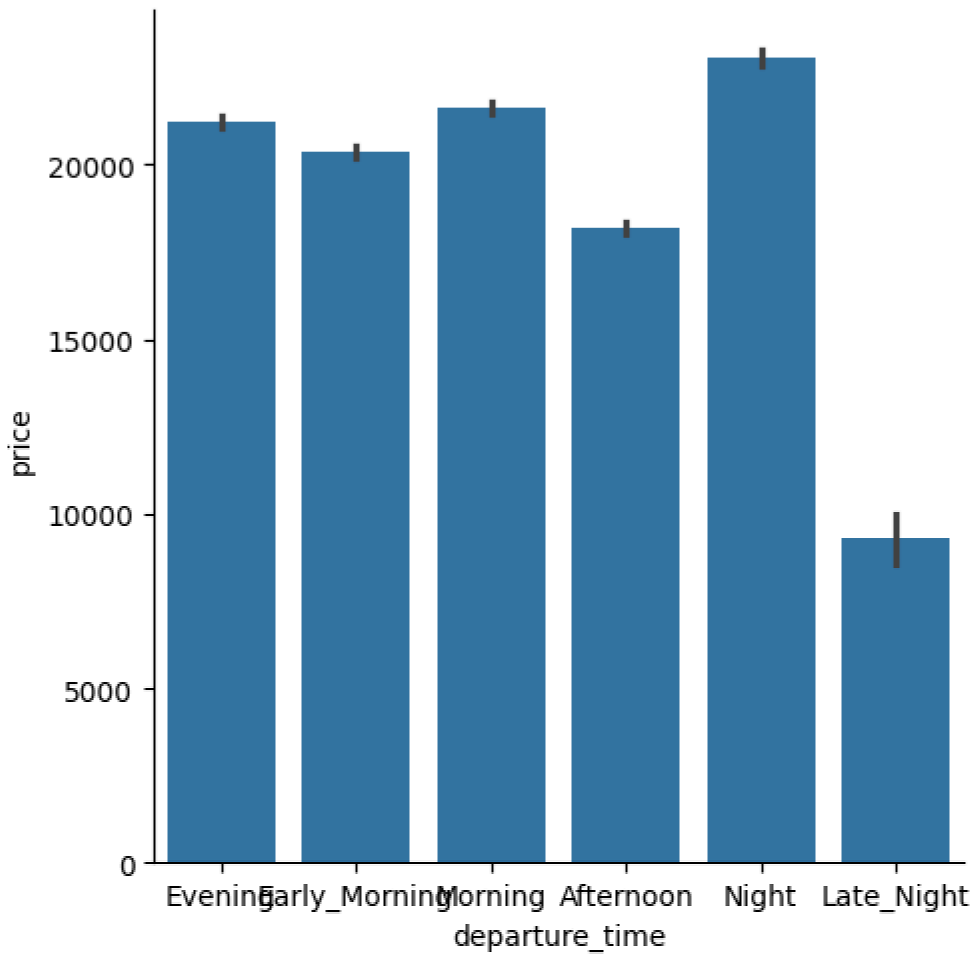
Out[93]:
```
departure_time
Afternoon        18179.203331
Early_Morning    20370.676718
Evening          21232.361894
Late_Night        9295.299387
Morning          21630.760254
Night            23062.146808
Name: price, dtype: float64
```

In [94]:
```
data.groupby('arrival_time')['price'].mean()
```

Out[94]:
```
arrival_time
Afternoon        18494.598993
Early_Morning    14993.139521
Evening          23044.371615
Late_Night       11284.906078
Morning          22231.076098
Night            21586.758341
Name: price, dtype: float64
```

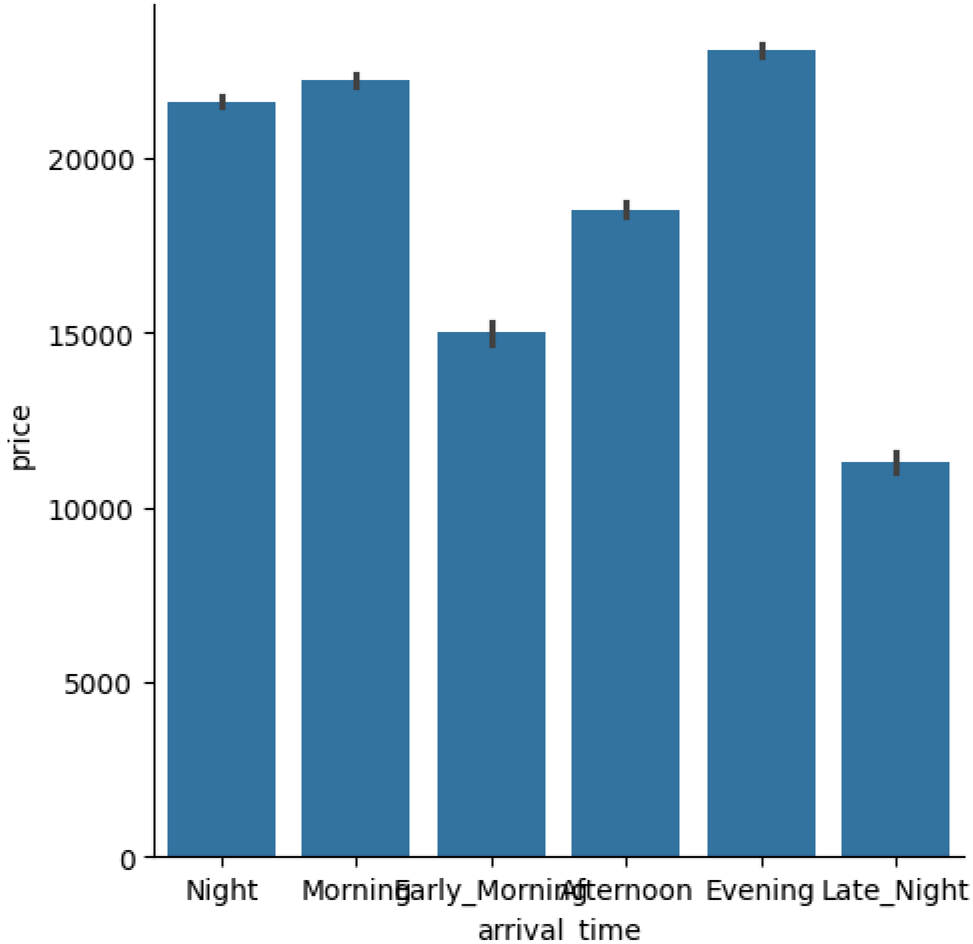
In [117…
```
plt.figure(figsize= (9,14))
sns.catplot(x = 'departure_time', y = 'price',data= data, kind = 'bar')
plt.show();
```

```
<Figure size 900x1400 with 0 Axes>
```



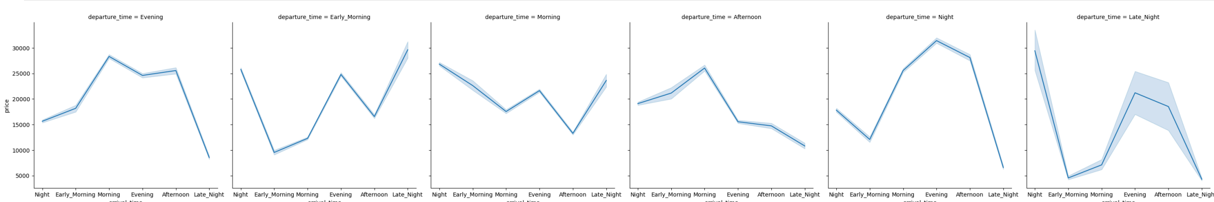
In [118…
```
plt.figure(figsize = (9,14))
sns.catplot(x = 'arrival_time', y = 'price',data= data, kind = 'bar')
plt.show();
```

```
<Figure size 900x1400 with 0 Axes>
```



In [119…

```
sns.relplot( x = 'arrival_time', y = 'price',data = data, col = 'departure_time', k
plt.show();
```



6. How prices changes with change in Source & Destination?

In [120…

```
data.groupby('source_city')['price'].mean()
```

Out[120…

```
source_city
Bangalore    21469.460575
Chennai      21995.339871
Delhi        18951.326639
Hyderabad    20155.623879
Kolkata      21746.235679
Mumbai       21483.818839
Name: price, dtype: float64
```

In [122…

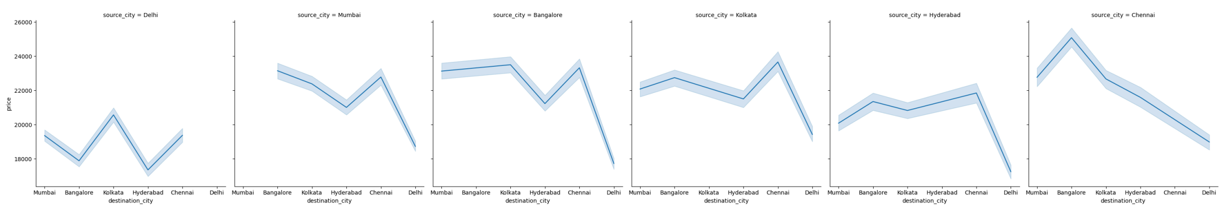
```
data.groupby('destination_city')['price'].mean()
```

Out[122…

```
destination_city
Bangalore    21593.955784
Chennai      21953.323969
Delhi        18436.767870
Hyderabad    20427.661284
Kolkata      21959.557556
Mumbai       21372.529469
Name: price, dtype: float64
```

In [123…

```
sns.relplot(x = 'destination_city', y  = 'price', data = data, col = 'source_city',
plt.show();
```



7. How is the Price affected when the Ticket is bought just 1 or 2 days before flight?

In [124…

```
data.head(2)
```

Out[124…

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city	cla
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai	Econor
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai	Econor

In [125…

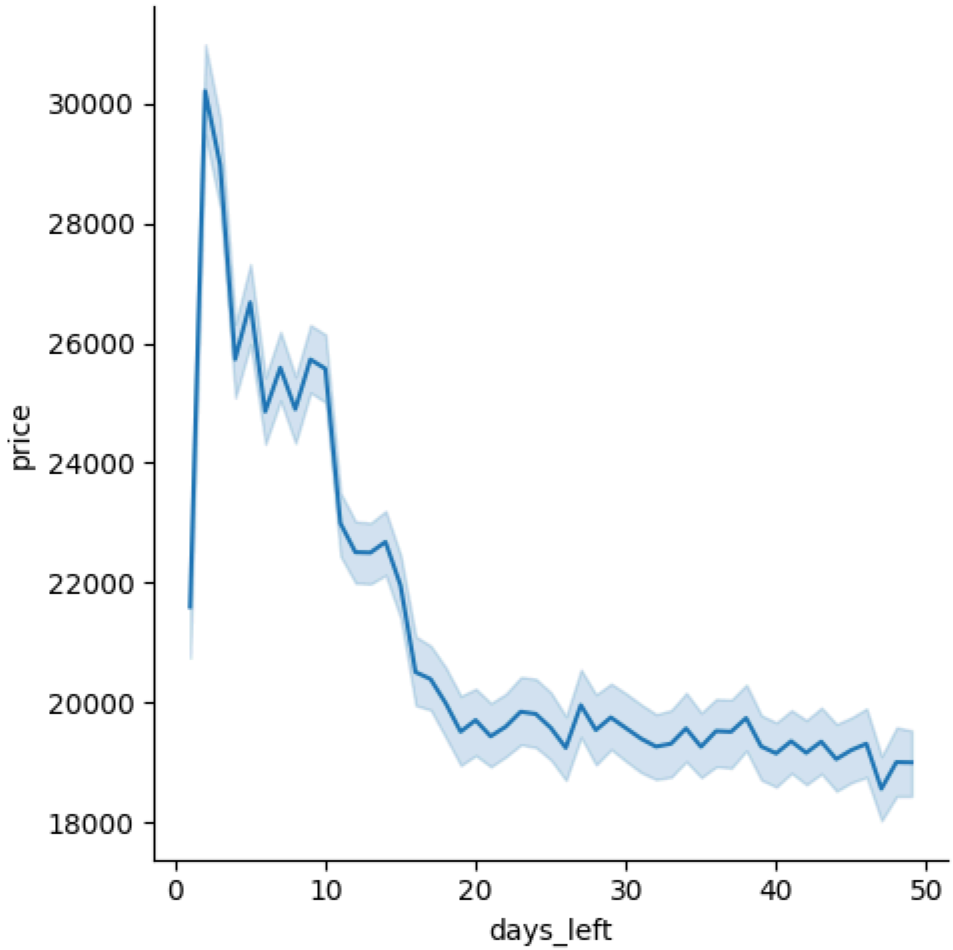
```
data.groupby('days_left')['price'].mean()
```

Out[125…

```
days_left
1     21591.867151
2     30211.299801
3     28976.083569
4     25730.905653
5     26679.773368
6     24856.493902
7     25588.367351
8     24895.883995
9     25726.246072
10    25572.819134
11    22990.656070
12    22505.803322
13    22498.885384
14    22678.002363
15    21952.540852
16    20503.546237
17    20386.353949
18    19987.445168
19    19507.677375
20    19699.983390
21    19430.494058
22    19590.667385
23    19840.913451
24    19803.908896
25    19571.641791
26    19238.290278
27    19950.866195
28    19534.986047
29    19744.653119
30    19567.580834
31    19392.706612
32    19258.135308
33    19306.271739
34    19562.008266
35    19255.652996
36    19517.688444
37    19506.306516
38    19734.912316
39    19262.095556
40    19144.972439
41    19347.440460
42    19154.261659
43    19340.528894
44    19049.080174
45    19199.876307
46    19305.351623
47    18553.272038
48    18998.126851
49    18992.971888
Name: price, dtype: float64
```

In [132…

```
sns.relplot(x = 'days_left', y = 'price', kind = 'line', data= data)
plt.show()
```



8. How Ticket Price varry in Economy and Business class?

In [133…

```
x = data[data['class'] == 'Economy']
x
```

Out[133…

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city
0	SpiceJet	SG-8709	Delhi	Evening	zero	Night	Mumbai
1	SpiceJet	SG-8157	Delhi	Early_Morning	zero	Morning	Mumbai
2	AirAsia	I5-764	Delhi	Early_Morning	zero	Early_Morning	Mumbai
3	Vistara	UK-995	Delhi	Morning	zero	Afternoon	Mumbai
4	Vistara	UK-963	Delhi	Morning	zero	Morning	Mumbai
...	...	...	...	...	...	...	...
206661	Vistara	UK-832	Chennai	Early_Morning	one	Night	Hyderabad
206662	Vistara	UK-832	Chennai	Early_Morning	one	Night	Hyderabad
206663	Vistara	UK-826	Chennai	Afternoon	one	Morning	Hyderabad
206664	Vistara	UK-822	Chennai	Morning	one	Morning	Hyderabad
206665	Vistara	UK-824	Chennai	Night	one	Night	Hyderabad

206666 rows × 11 columns

In [134…

```
x.price.mean()
```

Out[134…

```
np.float64(6572.342383362527)
```

In [135…

```
y = data[data['class'] == 'Business']
y
```

Out[135…

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city
206666	Air_India	AI-868	Delhi	Evening	zero	Evening	Mumbai
206667	Air_India	AI-624	Delhi	Evening	zero	Night	Mumbai
206668	Air_India	AI-531	Delhi	Evening	one	Night	Mumbai
206669	Air_India	AI-839	Delhi	Night	one	Night	Mumbai
206670	Air_India	AI-544	Delhi	Evening	one	Night	Mumbai
...	...	...	...	...	...	...	...
300148	Vistara	UK-822	Chennai	Morning	one	Evening	Hyderabad
300149	Vistara	UK-826	Chennai	Afternoon	one	Night	Hyderabad
300150	Vistara	UK-832	Chennai	Early_Morning	one	Night	Hyderabad
300151	Vistara	UK-828	Chennai	Early_Morning	one	Evening	Hyderabad
300152	Vistara	UK-822	Chennai	Morning	one	Evening	Hyderabad

93487 rows × 11 columns

In [136…

```
y.price.mean()
```

Out[136…

```
np.float64(52540.08112357868)
```

9. What will be the Average price of Vistara airline for a Flight from Dehli to Hyderabad in Business class?

In [140…

```
m = data[(data['airline'] == 'Vistara') & (data['class'] == 'Business') & (data['de
```

In [141…

```
m
```

Out[141…

	airline	flight	source_city	departure_time	stops	arrival_time	destination_city
219123	Vistara	UK-871	Delhi	Night	zero	Night	Hyderabad
219124	Vistara	UK-879	Delhi	Evening	zero	Evening	Hyderabad
219129	Vistara	UK-955	Delhi	Evening	one	Night	Hyderabad
219130	Vistara	UK-955	Delhi	Evening	one	Afternoon	Hyderabad
219131	Vistara	UK-985	Delhi	Evening	one	Night	Hyderabad
...	...	...	...	...	...	...	...
221863	Vistara	UK-963	Delhi	Morning	one	Early_Morning	Hyderabad
221864	Vistara	UK-985	Delhi	Evening	one	Early_Morning	Hyderabad
221865	Vistara	UK-985	Delhi	Evening	one	Afternoon	Hyderabad
221866	Vistara	UK-955	Delhi	Evening	one	Early_Morning	Hyderabad
221867	Vistara	UK-955	Delhi	Evening	one	Afternoon	Hyderabad

1660 rows × 11 columns

In [142…

```
m.price.mean()
```

Out[142…

```
np.float64(47939.840361445786)
```

In [ ]: